FACULTY OF ENGINEERING AND TECHNOLOGY

School of Computer Science and Engineering

(Computer Science & Engineering - CSE)

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A Project Report on

"Deepfake Detection and Blockchain Integration"

Submitted in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY IN

COMPUTER SCIENCE & ENGINEERING- CSE

Submitted by

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We would like to thank one and all who directly or indirectly helped us in the progress of our Project.

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ABSTRACT

This paper presents an extensive examination of a Deepfake Detection with Blockchain Verification framework. The project combines machine learning and blockchain technology to detect manipulated media and verify its authenticity in a secure and tamper-resistant manner. It explores the complete system architecture, including the deep learning model, data preprocessing, frame extraction, SHA-256 hashing, and blockchain verification through Solidity smart contracts, Python scripts, and Ethereum test networks such as Sepolia.

The paper also discusses recommended training procedures, suitable datasets, hyperparameter selection, model weight configuration, and implementation guidelines. Pseudo-code examples illustrate key operations such as extracting frames using OpenCV, generating predictions, computing hashes, and interacting with blockchain contracts using Web3.py. In addition, the study covers security considerations, testing strategies, CI/CD practices, and documentation methods to preserve originality.

The analysis is based on official resources from TensorFlow/Keras, OpenCV, Ethereum, Solidity, and Web3.py.

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Chapter 1
INTRODUCTION

Deepfake technology has emerged as one of the most advanced applications of Artificial Intelligence and Deep Learning in recent years. Deepfakes are manipulated images, videos, or audio generated using neural network architectures capable of producing highly realistic synthetic media. Such manipulated content closely resembles real media, making it difficult for humans to distinguish between authentic and fake content. The increasing availability of powerful AI tools and high-performance computing systems has significantly accelerated the development of deepfake generation techniques.

Initially, deepfake technology was mainly introduced for entertainment purposes such as movie production, animation, gaming, and virtual reality applications. Film industries use AI-generated facial replacement and voice synthesis for improving visual effects and reducing production costs. Similarly, educational and research organizations use synthetic media generation for simulation and virtual learning purposes. Although deepfake technology has several positive applications, its misuse has become a serious concern across the world.

Today, manipulated digital media is increasingly being used for spreading fake news, political misinformation, identity theft, cyberbullying, online fraud, fake evidence generation, and social manipulation. Deepfake videos of celebrities, politicians, and public figures are frequently circulated on social media platforms, creating confusion among people and damaging public trust in digital information.

Traditional methods used for media authentication are no longer sufficient because modern deepfakes contain realistic facial expressions, synchronized lip movement, natural lighting, and highly convincing visual details. Human observers often fail to identify manipulated content correctly. Therefore, intelligent automated systems capable of detecting fake media accurately are required.

The proposed project titled "Deepfake Detection and Blockchain Integration" aims to solve these challenges using a combination of Deep Learning and Blockchain Technology. The system uses the ResNet50 Convolutional Neural Network (CNN) model to classify uploaded media as REAL or FAKE. In addition, SHA-256 cryptographic hashing and Ethereum blockchain technology are integrated to securely store and verify media authenticity.

The proposed framework not only detects manipulated content but also ensures that uploaded media remains tamper-proof after verification. Blockchain integration provides immutable storage of media hashes, thereby improving digital trust and preventing unauthorized modifications.

The entire system is developed using TensorFlow/Keras for machine learning operations, OpenCV for image and video processing, Flask for backend services, and Web3.py for blockchain communication. HTML, CSS, and JavaScript are used for developing a responsive and user-friendly frontend interface.

The proposed solution aims to provide:

- Accurate deepfake detection
- Secure blockchain verification
- Tamper-proof digital authentication
- Improved digital trust
- Scalable and modular architecture
- Real-world applicability

The project contributes to research in Artificial Intelligence, Cybersecurity, Blockchain Technology, and Digital Forensics by demonstrating practical integration of modern technologies into a secure media verification framework.

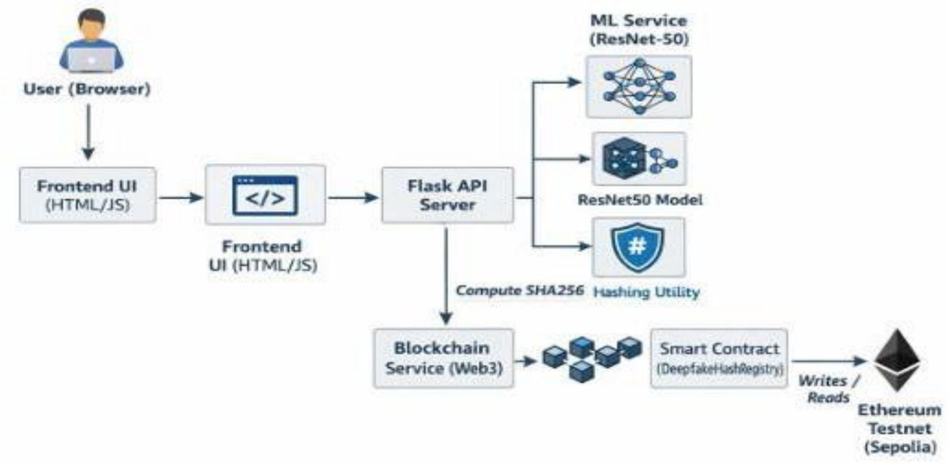


Figure 1: Architecture of the system (frontend communicates with the backend service that provides machine learning and blockchain services; the smart contract is on an Ethereum testnet).

1.1 Background & Motivation

The rapid growth of internet services and social media platforms has significantly increased digital communication and information sharing across the world. Millions of users upload photos, videos, and audio clips daily on online platforms such as YouTube, Instagram, Facebook, Twitter, and TikTok. Along with this digital growth, advancements in Artificial Intelligence and Deep Learning technologies have enabled the creation of highly realistic synthetic media known as deepfakes.
Deepfake generation techniques mainly rely on:

- Generative Adversarial Networks (GANs)
- Autoencoders
- Convolutional Neural Networks (CNNs)
- Variational Autoencoders (VAEs)

These neural network architectures analyze facial expressions, eye movement, voice patterns, and lip synchronization to generate manipulated media that appears highly realistic. The increasing accessibility of deepfake generation tools has made it possible even for non-technical users to create fake media easily.
Although deepfake technology has useful applications in:

- Entertainment
- Virtual reality
- Gaming
- Digital content creation
- Education

its misuse creates major ethical and security concerns.
Deepfakes can be used for:

- Fake political speeches
- Character assassination
- Financial fraud
- Identity theft
- Cyberbullying
- Fake legal evidence
- Social manipulation

One of the major concerns is that deepfake media spreads rapidly on social media platforms before verification can occur. This creates confusion among the public and damages trust in digital information systems.

Existing deepfake detection systems mainly focus only on classifying media as REAL or FAKE. However, they do not guarantee that verified media will remain unchanged in the future. After verification, files can still be manipulated or tampered with, making traditional authentication methods unreliable.

Blockchain technology offers a powerful solution to this problem because it provides:

- Decentralization
- Transparency
- Immutability
- Tamper-proof storage
- Secure verification

By storing SHA-256 hashes of uploaded media on blockchain networks, it becomes possible to verify whether media files have been modified after authentication

The motivation behind this project is to integrate Deep Learning and Blockchain Technology to create a secure and intelligent framework capable of:

- Detecting manipulated media
- Securing media authenticity
- Preventing tampering
- Improving digital trust

The project also aims to support applications in cybersecurity, journalism, banking, social media monitoring, and digital forensics.

1.2 Objectives

The primary objective of the proposed project is to develop an intelligent and secure system capable of detecting deepfake media and verifying digital authenticity using blockchain technology.

The specific objectives of the project are described below.

Deepfake Detection

To accurately classify uploaded images and videos into:

- REAL
- FAKE

using Deep Learning techniques and CNN architectures.

The system uses ResNet50 transfer learning architecture to improve feature extraction and prediction accuracy.

Blockchain-Based Verification

To securely store SHA-256 hash values of uploaded media files on Ethereum blockchain using Solidity smart contracts.

Hash Formula:

$$= 256()$$

Where:

- H = Generated Hash
- M = Media File

This ensures immutable and tamper-proof verification of digital media.

Secure Digital Authentication

To verify whether uploaded media files remain unchanged after authentication.

The system compares newly generated hashes with blockchain records to identify modifications or tampering.

User-Friendly Interface

To develop a simple and responsive web application that allows users to:

- Upload media
- View prediction results
- Check confidence scores
- Verify blockchain authenticity

Improve Detection Accuracy

To improve deepfake detection performance using:

• Transfer learning
• Data preprocessing
• Data augmentation
• Fine-tuning techniques

Reduce Digital Misinformation

To prevent the spread of fake media and manipulated information across social media and online platforms.

Provide Scalable Architecture

To develop a modular system architecture that supports future improvements such as:

- Real-time detection
- Audio deepfake detection
- Mobile application deployment
- Cloud integration

Support Cybersecurity Applications

To assist in:

- Digital forensics
- Law enforcement
- Banking security
- Media verification
- Cybercrime investigation

The proposed system aims to improve trust and security in digital communication systems.

1.3 Benefits of Research

The proposed research provides several advantages in the field of Deepfake Detection, Artificial Intelligence, Cybersecurity, and Blockchain Technology.

Improved Deepfake Detection Accuracy

The ResNet50 CNN model improves classification performance by extracting complex visual and facial features from images and video frames.

Benefits include:

- Better feature extraction
- High prediction accuracy
- Improved model generalization
- Reduced false predictions

The use of transfer learning also reduces training time and improves efficiency.

Secure Blockchain Verification

Blockchain technology ensures secure and immutable verification of uploaded media.

Advantages include:

- Tamper-proof storage
- Decentralized verification

- Transparent authentication
- Immutable record maintenance

The blockchain ensures that media authenticity records cannot be modified after storage.

Prevention of Digital Fraud

The proposed system helps prevent:

- Identity theft
- Fake news generation
- Financial fraud
- Fake evidence creation
- Social manipulation
- Cybercrime activities

This improves security and trust in digital environments.

Improved Digital Trust

The integration of blockchain technology improves trust in:

- Online communication
- Social media platforms
- Journalism
- Digital transactions
- Public information systems

Users can verify whether digital media is authentic before trusting or sharing it.

Scalability and Flexibility

The modular architecture allows future expansion and scalability.
Possible future enhancements include:

- Real-time video detection
- Audio deepfake detection
- Multi-blockchain integration
- Cloud deployment
- Mobile applications

Academic and Research Contribution

The project contributes to research areas such as:

- Artificial Intelligence
- Deep Learning
- Blockchain Technology
- Cybersecurity
- Digital Forensics

It demonstrates practical integration of modern technologies into a real-world security application.

CHAPTER 2
Literature Survey

Deepfake detection has become an important research area because of the rapid increase in manipulated digital media and AI-generated content. Researchers across the world are developing different methods using Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, and Blockchain Technology for detecting fake media and ensuring digital authenticity.

The literature survey helps in understanding:

- Existing technologies
- Deepfake detection methods
- CNN architectures
- Blockchain verification systems
- Cryptographic hashing techniques
- Research limitations and gaps

This chapter reviews several research works related to deepfake generation, deepfake detection, blockchain verification, and digital media authentication.

Modern deepfake generation methods use advanced neural networks capable of generating highly realistic synthetic media. As deepfake quality improves, traditional detection systems become less effective.

Therefore, researchers are increasingly focusing on developing intelligent detection systems capable of identifying subtle inconsistencies present in manipulated media.

Similarly, blockchain technology is gaining importance because of its decentralized and tamper-proof nature. Researchers are integrating blockchain with digital authentication systems to improve media security and trustworthiness.

The following sections discuss various existing research works and techniques used in deepfake detection and blockchain-based verification systems.

CHAPTER 3
System Analysis and Proposed Methodology

3.1 Introduction

Deepfake technology has become one of the major challenges in digital communication systems due to rapid advancements in Artificial Intelligence and Deep Learning. Modern deepfake generation techniques are capable of producing highly realistic manipulated images and videos that are difficult to distinguish from genuine media. These fake media files can be used for misinformation, cybercrime, identity theft, financial fraud, political manipulation, and social engineering attacks.

Existing deepfake detection systems mainly focus only on identifying whether media content is real or fake. However, most systems do not provide secure mechanisms for verifying media authenticity after detection. Even after successful verification, digital media can still be modified or tampered with, creating security concerns.

To overcome these problems, the proposed project introduces a Deepfake Detection and Blockchain Integration system that combines Deep Learning and Blockchain Technology into a unified framework. The proposed system uses ResNet50-based Convolutional Neural Network (CNN) architecture for deepfake detection and Ethereum blockchain technology for secure media verification.

The system analyzes uploaded images and videos to classify them as REAL or FAKE. Simultaneously, SHA-256 hash values of uploaded media files are generated and securely stored on the blockchain using Solidity smart contracts. Blockchain integration ensures tamper-proof verification and secure digital authentication.

The proposed methodology provides:

- Accurate deepfake detection
- Secure blockchain verification
- Tamper-proof authentication
- Scalable architecture
- User-friendly interaction

The following sections explain the problem statement, architecture, algorithms, and implementation methodology in detail.

3.2 Problem Statement

The rapid growth of Artificial Intelligence and Deep Learning technologies has significantly increased the generation of manipulated media known as deepfakes. Modern deepfake generation tools can create highly realistic fake images and videos using neural networks such as GANs and Autoencoders. These manipulated media files are difficult to distinguish from original content because they contain realistic facial expressions, synchronized lip movement, and natural visual appearance.

Deepfakes create serious threats in multiple areas such as:

- Fake news generation
- Political misinformation
- Financial fraud
- Identity theft
- Social manipulation
- Cybercrime
- Fake evidence generation

Traditional media verification systems are no longer sufficient because human observers often fail to identify manipulated content accurately. Existing deepfake detection systems also suffer from several limitations such as:

- Lack of secure verification
- No tamper-proof storage
- Poor scalability
- High computational cost
- Limited real-world performance
- Vulnerability to adversarial attacks

Most existing systems only classify media into REAL or FAKE categories without ensuring that the media remains authentic after verification. Files can still be modified after authentication, making traditional verification methods unreliable.

Therefore, there is a need for a secure and intelligent framework capable of:

1. Detecting manipulated media accurately
2. Preventing unauthorized tampering
3. Providing immutable authenticity records
4. Supporting secure digital verification

The proposed system addresses these challenges by integrating:

- ResNet50 Deepfake Detection
- SHA-256 Cryptographic Hashing
- Ethereum Blockchain Verification

This integration improves digital trust and ensures secure media authentication.

3.3 System Architecture / Model

The proposed system architecture consists of four major modules:

1. Frontend Module
2. Backend Module
3. Machine Learning Module
4. Blockchain Module

Each module performs specific operations while communicating with other components of the system.

Frontend Module

The frontend provides interaction between users and the system.

Technologies Used:

- HTML
- CSS
- JavaScript

Functions:

- Media upload
- Display prediction results
- Show confidence scores
- Verify blockchain authenticity

The frontend interface is designed to be responsive, interactive, and user-friendly.

Backend Module

The backend acts as the central controller of the system.

Technology Used:

- Flask Framework

Functions:

- Handle API requests
- Manage uploaded files
- Connect frontend with ML model
- Communicate with blockchain network
- Generate JSON responses

The backend ensures smooth communication between all system modules.

Machine Learning Module

The Machine Learning module performs deepfake detection using ResNet50 CNN architecture.

Technologies Used:

- TensorFlow
- Keras
- OpenCV

Functions:

- Image preprocessing
- Feature extraction
- Classification
- Confidence score generation

The system processes:

- Images directly
- Video frames extracted using OpenCV

The output generated is:

- REAL
- FAKE

along with prediction confidence.

Blockchain Module

The blockchain module ensures secure verification and tamper-proof storage.

Technologies Used:

- Ethereum Blockchain
- Solidity Smart Contracts
- Web3.py
- Sepolia Testnet

Functions:

- Generate SHA-256 hash
- Store media hashes on blockchain
- Verify media authenticity
- Prevent unauthorized tampering

The blockchain stores only hash values rather than complete media files, improving efficiency and reducing storage cost.

Workflow of Proposed System

The complete workflow is as follows:

1. User uploads image or video
2. Backend receives uploaded file
3. Preprocessing is performed
4. Frames extracted if video
5. ResNet50 predicts REAL or FAKE
6. SHA-256 hash generated
7. Hash stored on blockchain
8. Prediction results displayed
9. Future verification performed using blockchain records

The modular architecture improves scalability, security, and maintainability.

3.4 Proposed Algorithms

The proposed system uses multiple algorithms and techniques for deepfake detection and secure verification.

3.4.1 ResNet50 Algorithm

ResNet50 is a deep Convolutional Neural Network containing 50 layers.

The model uses residual learning to solve the vanishing gradient problem.

Residual Formula:

$$() = () +$$

Where:

- H(x) = Output
- F(x) = Residual mapping
- x = Input

Advantages of ResNet50

- High classification accuracy
- Better feature extraction
- Improved deep learning performance
- Supports transfer learning

3.4.2 SHA-256 Hashing Algorithm

SHA-256 is a cryptographic hashing algorithm used for secure media authentication.

Hash Representation:

$$= 256()$$

Where:

- H = Generated Hash
- M = Input Media File

Features

- One-way hashing

- Collision resistant
- Secure digital fingerprint
- Fixed output length

SHA-256 ensures media integrity verification.

3.4.3 Blockchain Smart Contract Algorithm

The blockchain module uses Solidity smart contracts.

Main Functions:

- storeHash()
- verifyHash()

Workflow

1. Generate hash value
2. Convert to blockchain-compatible format
3. Store on Ethereum blockchain
4. Verify whenever required

Blockchain ensures immutable and transparent verification.

3.5 Proposed Work

The proposed work focuses on developing a secure and intelligent framework for deepfake detection and blockchain verification.

The implementation integrates:

- Deep Learning
- Blockchain Technology
- Cryptographic Hashing
- Web Technologies

The proposed work includes the following stages.

Dataset Collection

Datasets used include:

- FaceForensics++
- Celeb-DF
- DFDC Dataset

These datasets contain both real and manipulated media used for model training and testing.

Data Preprocessing

Preprocessing techniques include:

- Frame extraction
- Face detection
- Image resizing
- Normalization
- Data augmentation

These operations improve model performance and training efficiency.

Model Training

The ResNet50 model is trained using transfer learning.

Training includes:

- Loading pretrained weights
- Freezing base layers
- Adding custom dense layers
- Fine-tuning selected layers

The model predicts whether uploaded media is:

- REAL
- FAKE

Hash Generation

SHA-256 hashing is used to generate unique digital fingerprints for uploaded media files.

The generated hash ensures:

- Integrity verification
- Tamper detection
- Secure authentication

Blockchain Integration

Generated hashes are securely stored on Ethereum blockchain using Solidity smart contracts.

The blockchain provides:

- Immutable storage
- Secure verification
- Decentralized authentication

Frontend and Backend Development

The frontend is developed using:

- HTML
- CSS
- JavaScript

The backend uses:

- Flask
- TensorFlow
- Web3.py

API Endpoints include:

- /upload
- /verify
- /health

Security Features

The system includes:

- SHA-256 hashing

- Blockchain immutability
- Secure file upload
- API validation
- Secure blockchain communication

Expected Outcome

The proposed system aims to achieve:

- High deepfake detection accuracy
- Secure media verification
- Tamper-proof authentication
- Improved digital trust
- Scalable deployment

The developed framework can be used in:

- Journalism
- Banking
- Cybersecurity
- Digital forensics
- Social media monitoring
- Law enforcement systems

CHAPTER 4
System Design and Implementation

4.1 Introduction

System design and implementation play an important role in developing reliable and efficient software applications. The proposed Deepfake Detection and Blockchain Integration system is designed using modular architecture to ensure scalability, maintainability, and secure communication between components.

The system integrates:

- Deep Learning
- Blockchain Technology
- Cryptographic Hashing
- Web Technologies

The implementation focuses on:

- Deepfake detection
- Secure blockchain verification
- User-friendly interaction
- Efficient media processing

The following sections explain the system design principles, architecture, dataset preprocessing, model implementation, application design, and implementation workflow.

4.2 System Design Principles

The proposed system follows multiple design principles to ensure efficient performance and security.

Modular Architecture

The system is divided into independent modules:

- Frontend
- Backend
- Machine Learning
- Blockchain

This improves maintainability and scalability.

Scalability
The architecture supports future enhancements such as:

- Real-time detection
- Audio deepfake analysis
- Mobile deployment
- Cloud integration

Security
Security measures include:

- SHA-256 hashing
- Blockchain verification
- Secure API communication
- Input validation

Efficiency

Efficient algorithms and preprocessing techniques reduce computational complexity and improve prediction speed.

User-Friendliness

The interface is designed to provide:

- Easy media upload
- Real-time prediction display
- Simple verification process

4.3 System Architecture

The proposed architecture consists of four major layers:

1. User Interface Layer
2. Application Layer
3. Machine Learning Layer
4. Blockchain Layer

The user uploads media files through the frontend interface. The backend processes uploaded files and communicates with both the machine learning and blockchain modules.

The machine learning module performs deepfake detection using ResNet50, while the blockchain module securely stores and verifies media hashes.

The architecture ensures:

- Secure communication
- Fast prediction
- Tamper-proof verification
- Scalable deployment

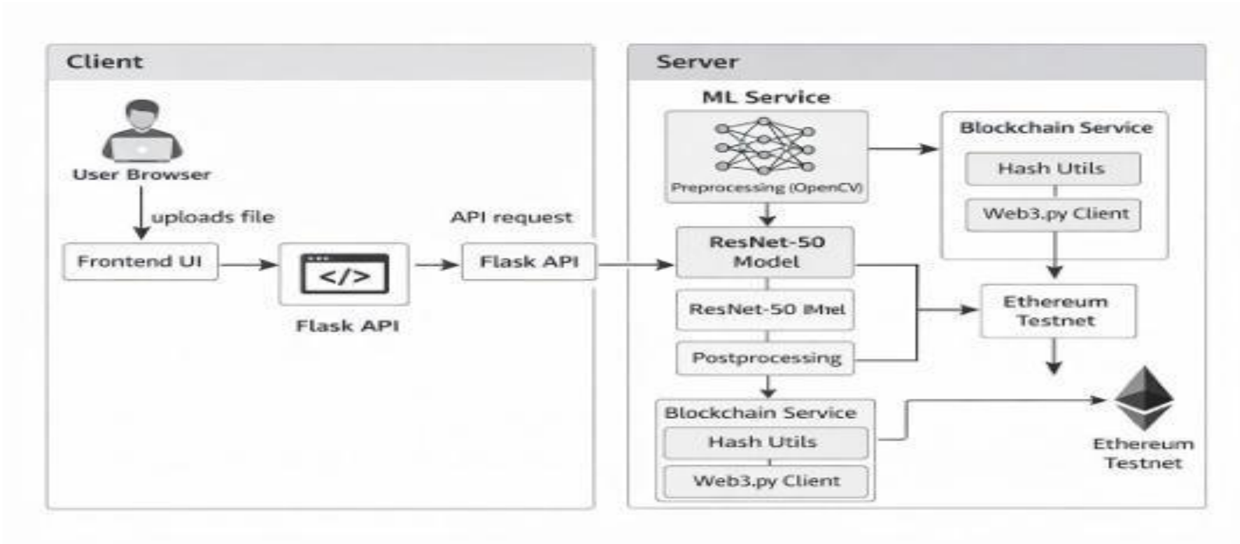


Figure 2: Schematic representation of components (routes from flask application are redirected to ML and Blockchain microservices; ML service contains pre-processing and inference; Blockchain microservice contains hashing and web3 calls).

4.4 Dataset Design & Preprocessing

The quality of datasets significantly affects model performance.

Datasets used:

- FaceForensics++
- Celeb-DF
- DFDC Dataset

These datasets contain:

- Real images
- Fake images
- Manipulated videos

4.4.1 Dataset Structure

The dataset is organized into:

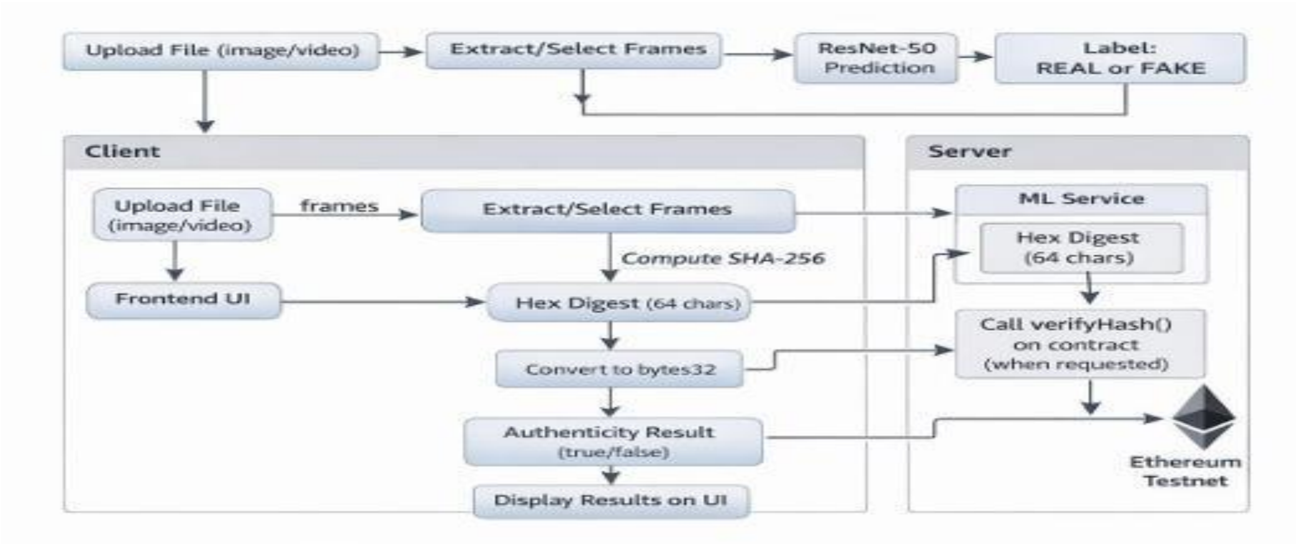
- Training set
- Validation set
- Testing set

Typical split:

- Training: 80%
- Validation: 10%
- Testing: 10%

This improves model generalization and performance evaluation.

Data flow Diagram Figure



3: Data flow for detection (left) and blockchain validation (right). The upload of the file initiates ML prediction and hashing. The hash is uploaded to the blockchain (detect), and then the validation process checks whether the file is genuine.

4.4.2 Preprocessing Techniques

Preprocessing includes:

- Frame extraction

- Face detection
- Image resizing
- Normalization

Images are resized to:
224 × 224 pixels

Normalization Formula:

$$_{omlzd} = 255$$

Preprocessing improves:

- Feature extraction
- Training efficiency
- Prediction accuracy

4.4.3 Data Augmentation

Data augmentation artificially increases dataset diversity.

Techniques include:

- Rotation
- Flipping
- Zooming
- Cropping
- Brightness adjustment

Benefits:

- Reduces overfitting
- Improves robustness
- Enhances generalization

4.5 Model Architecture & Implementation

The system uses ResNet50 CNN architecture for deepfake detection.

The model is implemented using:

- TensorFlow
- Keras

The architecture includes:

- Convolution layers
- Pooling layers
- Residual blocks
- Dense layers

Transfer learning improves model performance and reduces training time.

CHAPTER 5

Tools and Technologies Required

5.1 Development Tools

The proposed Deepfake Detection and Blockchain Integration system requires several software tools and development platforms for implementing machine learning operations, blockchain communication, frontend design, backend services, testing, and deployment. These tools help in building a secure, scalable, and efficient application.

Python

Python is the primary programming language used for implementing the proposed system. It is widely used in Artificial Intelligence, Machine Learning, and Blockchain development because of its simplicity and large collection of libraries.

Python is used for:

- Building machine learning models
- Performing image and video preprocessing
- Developing backend APIs
- Integrating blockchain services
- Generating SHA-256 hashes

The major advantage of Python is its easy syntax and strong support for AI and data science applications.

Visual Studio Code

Visual Studio Code (VS Code) is used as the main development environment for coding and debugging the project.
It provides:

- Syntax highlighting
- Integrated terminal
- Debugging support
- Extension support
- Version control integration

VS Code improves productivity and simplifies project management during development.

Jupyter Notebook

Jupyter Notebook is used during model training and experimentation.

It helps in:

- Data analysis
- Visualization
- Model testing
- Performance evaluation

Jupyter Notebook allows interactive execution of code and makes debugging easier during machine learning implementation.

Git and GitHub

Git is used for version control while GitHub is used for maintaining online repositories.

These tools help in:

- Tracking code changes
- Managing project versions
- Collaborating with developers
- Creating project backups

GitHub also helps in maintaining proper documentation and project organization.

OpenCV

OpenCV is an open-source computer vision library used for image and video processing.

It is used for:

- Frame extraction from videos
- Face detection
- Image preprocessing
- Video analysis
- Image resizing and normalization

OpenCV improves media processing efficiency and supports deepfake detection operations.

5.2 Frontend Technologies

Frontend technologies are used for developing the user interface of the system. The frontend enables users to upload media files, view prediction results, and verify blockchain authenticity.

HTML

HTML (HyperText Markup Language) is used for designing the structure of web pages.

It is responsible for:

- Creating upload forms
- Designing result sections
- Displaying verification outputs
- Building page layouts

HTML forms the foundation of the frontend interface.

CSS

CSS (Cascading Style Sheets) is used for styling and improving the visual appearance of the application.

CSS helps in:

- Responsive web design
- Color management
- Font styling
- Layout arrangement
- Animation effects

A properly designed CSS interface improves user experience and application usability.

JavaScript

JavaScript is used for implementing dynamic and interactive frontend functionality.

It performs:

- Form validation
- Real-time updates
- API communication
- Dynamic content rendering
- User interaction handling

JavaScript improves frontend responsiveness and overall system interaction.

Bootstrap

Bootstrap is used for developing responsive and mobile-friendly user interfaces.

Bootstrap provides:

- Predefined UI components
- Responsive grid system
- Faster frontend development
- Cross-browser compatibility

The use of Bootstrap ensures that the application works properly across different devices and screen sizes.

5.3 Backend Technologies

Backend technologies are responsible for processing uploaded media, communicating with machine learning models, and interacting with blockchain networks.

Flask Framework

Flask is a lightweight Python web framework used for backend development.

Flask is responsible for:

- Handling API requests
- Managing uploaded files
- Connecting frontend and backend
- Returning prediction results
- Communicating with blockchain services

Flask is simple, flexible, and highly suitable for machine learning-based web applications.

TensorFlow

TensorFlow is an open-source deep learning framework developed by Google.

It is used for:

- Training deep learning models
- Building CNN architectures

- Running prediction operations
- GPU-accelerated computation

TensorFlow provides strong support for large-scale machine learning operations.

Keras

Keras is a high-level deep learning API built on top of TensorFlow.

Keras simplifies:

- Neural network implementation
- Layer management
- Model compilation
- Training configuration

It allows faster and easier implementation of deep learning architectures.

Web3.py

Web3.py is a Python library used for communication with Ethereum blockchain.

It performs:

- Smart contract interaction
- Blockchain transaction handling
- Hash verification
- Reading blockchain data

Web3.py acts as a bridge between Python applications and blockchain networks.

Solidity

Solidity is the programming language used for developing Ethereum smart contracts.

The smart contracts are responsible for:

- Storing media hashes
- Verifying authenticity
- Managing blockchain operations

Solidity provides secure and decentralized execution of blockchain functions.

5.4 Deployment & Hosting Tools

Deployment tools are used for executing and hosting the proposed application.

Ethereum Sepolia Testnet

The Sepolia Testnet is used for testing Ethereum smart contracts.

It provides:

- Free blockchain testing
- Secure development environment
- Transaction validation
- Smart contract deployment

The testnet allows developers to test blockchain functionality without using real cryptocurrency.

Ganache

Ganache is a local blockchain simulator used during development and testing.

Ganache helps in:

- Local blockchain deployment
- Smart contract testing
- Transaction debugging
- Faster blockchain experimentation

It simplifies blockchain development before deploying to public networks.

MetaMask

MetaMask is a cryptocurrency wallet used for Ethereum transactions.

It performs:

- Wallet management
- Transaction signing
- Smart contract interaction
- Blockchain authentication

MetaMask provides secure blockchain communication between the application and Ethereum network.

Flask Deployment

The backend application can be deployed using Flask servers.

Flask deployment provides:

- Real-time API access
- Backend execution
- Web application hosting
- Scalable deployment support

5.5 Testing & Monitoring Tools

Testing tools are used to ensure system reliability, security, and performance.

PyTest

PyTest is used for backend testing and validation.

It helps in:

- Unit testing
- Functional testing
- API testing
- Integration testing

PyTest ensures that backend operations work correctly.

Postman

Postman is used for testing REST APIs developed using Flask.

It allows developers to:

- Send API requests
- Validate responses
- Test endpoints
- Debug backend communication

Postman improves API reliability and debugging efficiency.

TensorBoard

TensorBoard is used for monitoring deep learning model training.

It provides:

- Accuracy graphs
- Loss visualization
- Performance tracking
- Training analytics

TensorBoard helps in evaluating model behavior during training.

Logging Tools

Logging tools are used for monitoring system activities.

Logs help in tracking:

- Errors
- Predictions
- Transactions
- API requests
- Blockchain communication

Logging improves debugging and system maintenance.

5.6 Optional / Future Integration Tools

Future improvements may require additional tools and technologies.

Cloud Platforms

Cloud services such as AWS, Google Cloud, and Microsoft Azure may be used for:

- Cloud deployment
- Scalable storage
- Distributed processing
- Real-time detection systems

Cloud integration improves scalability and accessibility.

Mobile Application Frameworks

Frameworks such as Flutter and React Native may be used for mobile application development.

Mobile support can provide:

- Easy accessibility
- Real-time verification
- Portable deepfake detection

Real-Time Streaming Tools

Technologies such as Kafka and WebSockets may be integrated for:

- Live video analysis
- Real-time deepfake detection
- Streaming verification systems

Advanced AI Models

Future systems may use advanced architectures such as:

- EfficientNet
- Vision Transformers (ViT)
- Swin Transformer

These models may further improve deepfake detection accuracy and performance.

CHAPTER 6

Software Requirement Specification (SRS)

6.1 Specific Requirements

The Software Requirement Specification (SRS) defines the functional and non-functional requirements of the proposed Deepfake Detection and Blockchain Integration system. The purpose of this document is to describe the system behavior, user interaction, software interfaces, communication mechanisms, and performance requirements necessary for successful implementation.

The proposed system is designed to detect manipulated media using Deep Learning techniques and securely verify media authenticity using Blockchain Technology. The system should support image and video uploads, perform deepfake detection efficiently, generate SHA-256 hashes, and store verification records securely on Ethereum blockchain.

The software requirements include:

- User-friendly web interface
- Secure backend communication
- Efficient machine learning prediction
- Blockchain integration support
- Real-time response generation
- Secure file processing and verification

The system must also provide high accuracy, reliability, scalability, and maintainability for real-world deployment.

Model Weights Options

Option	Description	Pros	Cons	Recommenddion
Pretrained (ImageNet)	Initialize ResNet50 with ImageNet weights, fine-tune on deepfake data.	Significantly reduces training time; leverages general features. Good performance reported (~90%–97% accuracy on FaceForensics+, CelebDF).	May need careful fine-tuning to adapt to new domain; slight risk of domain mismatch.	Generally recommended, especially if dataset is moder-ate. Pretrained base + custom top layers.
From Scratch	Randomly initialize ResNet50 (or smaller CNN) and train on deepfake dataset.	No reliance on external dataset; potentially high accuracy if dataset is massive.	Requires very large labeled dataset and compute; likely slow convergence; risk of overfitting if data is limited.	Only if you have thousands of deepfake examples and GPU resources. Otherwise use transfer learning.

Table 1: Model weights approaches comparison table. Considering the real-life limitations, the use of transfer learning (pre-trained ResNet model) would be better. That's because in some studies published, for example, ResNet50 pre-trained using CelebDF and FaceForensics++ datasets obtained an accuracy of 97%, while others CNN models used only FaceForensics++ dataset scored just 81%.

6.2 Functional Requirements

Functional requirements describe the operations and services provided by the system.

Media Upload Functionality

The system should allow users to upload:

- Image files
- Video files

Supported formats include:

- JPG
- PNG
- JPEG
- MP4
- AVI

The uploaded files should be validated before processing to prevent unsupported formats and corrupted media.

Deepfake Detection Functionality

The system should analyze uploaded media using the ResNet50 CNN model and classify the content as:

- REAL
- FAKE

The model should process:

- Images directly
- Video frames extracted using OpenCV

The prediction results should be displayed clearly to the user.

Confidence Score Generation

The system should display confidence scores representing prediction certainty.

Example:

- REAL – 96%
- FAKE – 93%

Confidence scores help users understand prediction reliability.

SHA-256 Hash Generation

The system should generate SHA-256 hash values for uploaded media files.

Hash Formula:

$$= 256()$$

Where:

- H = Generated Hash
- M = Media File

The generated hash acts as a unique digital fingerprint for integrity verification.

Blockchain Storage Functionality

The system should securely store generated hash values on Ethereum blockchain using Solidity smart contracts.

The blockchain storage mechanism should ensure:

- Tamper-proof verification
- Immutable record maintenance
- Secure authentication

Blockchain Verification Functionality

The system should verify uploaded media authenticity by comparing newly generated hashes with blockchain records.

If hashes match:

- The file is authentic

If hashes do not match:

- The file has been modified or tampered with

API Management Functionality

The backend should provide REST APIs for frontend communication.

Important API endpoints include:

- /upload

- /verify
- /health

These APIs should support secure data transmission and efficient response handling.

User Interface Functionality

The frontend interface should provide:

- Media upload options
- Prediction result display
- Confidence score visualization
- Verification status display
- Error notifications

The interface should be simple and easy to use.

Logging and Monitoring Functionality

The system should maintain logs for:

- Upload activities
- Prediction results
- Blockchain transactions
- Error handling
- API requests

Logs improve debugging and system monitoring.

6.3 Non-Functional Requirements

Non-functional requirements define the quality, reliability, performance, and usability of the system.

Performance Requirements

The system should:

- Process uploaded media efficiently
- Generate predictions within acceptable response time

- Handle multiple user requests simultaneously

The prediction system should maintain fast response time for both image and video processing.

Security Requirements

The system should provide strong security mechanisms such as:

- Secure API communication
- Blockchain-based verification
- SHA-256 hashing
- Input validation
- Secure file management

Private blockchain keys must be protected to prevent unauthorized access.

Scalability Requirements

The architecture should support:

- Future feature expansion
- Increased user traffic
- Large-scale dataset handling
- Real-time detection integration

The modular design improves scalability and maintainability.

Reliability Requirements

The system should provide:

- Stable predictions
- Accurate classification
- Reliable blockchain communication
- Consistent verification results

The application should minimize prediction failures and processing errors.

Availability Requirements

The system should remain available for continuous user interaction and blockchain verification.

Cloud deployment can improve availability and uptime in future implementations.

Maintainability Requirements

The system should support:

- Easy debugging
- Modular updates
- Future improvements
- Code maintainability

Proper documentation and modular coding improve system maintenance.

6.4 Human-Centered Design Requirements

Human-centered design focuses on improving user experience, accessibility, and usability.

User-Friendly Interface

The application interface should be simple, interactive, and easy to understand.

Users should be able to:

- Upload files easily
- View results clearly
- Verify media authenticity quickly

The design should minimize user confusion.

Responsive Design

The application should support:

- Desktop systems
- Mobile devices
- Tablets

Responsive web design improves accessibility across multiple platforms.

Accessibility Requirements

The interface should use:

- Readable fonts
- Proper spacing
- Clear buttons
- Good color contrast

This improves usability for different types of users.

Error Handling and Notifications

The system should provide clear error messages during:

- Invalid file uploads
- Blockchain failures
- API communication errors
- Unsupported formats

Proper error notifications improve user experience.

Easy Navigation

The frontend should provide smooth navigation between:

- Upload page
- Prediction results
- Verification sections

Users should easily understand the workflow without technical knowledge.

6.5 Specific Requirements

The proposed system requires:

- Python runtime environment
- GPU support for model training

- Internet connection for blockchain communication
- Ethereum wallet integration
- Modern web browser support

The software should support secure blockchain interaction and machine learning prediction.

6.5.1 Software Interfaces

Software interfaces define communication between system components.

Frontend Interface

The frontend interface communicates with backend APIs using HTTP requests.

Functions include:

- Media upload
- Prediction requests
- Verification requests
- Result visualization

Backend Interface

The backend connects:

- Frontend services
- Machine learning model
- Blockchain network

The backend handles API routing and data processing.

Machine Learning Interface

The ML interface connects the trained ResNet50 model with backend services.

Functions include:

- Image preprocessing
- Prediction generation
- Confidence score calculation

Blockchain Interface

The blockchain interface uses Web3.py to communicate with Ethereum smart contracts.

Functions include:

- Hash storage
- Blockchain transactions
- Hash verification
- Reading blockchain records

6.5.2 Communication Interfaces

Communication interfaces define data exchange between system modules.

Frontend and Backend Communication

Communication occurs using:

- HTTP protocol
- REST APIs
- JSON responses

The frontend sends media files and receives prediction results from backend services.

Backend and Machine Learning Communication

The backend sends preprocessed media data to the machine learning module and receives prediction outputs.

The communication includes:

- Image arrays
- Video frames
- Prediction labels
- Confidence scores

Backend and Blockchain Communication

The backend communicates with Ethereum blockchain using Web3.py.

Functions include:

- Smart contract execution
- Hash storage
- Verification requests
- Blockchain transaction management

Security Communication

Secure communication mechanisms include:

- Input validation
- Protected blockchain credentials
- Secure API requests
- Controlled transaction execution

These mechanisms improve system reliability and cybersecurity.

CHAPTER 7
Results and Discussion

7.1 Introduction

This chapter presents the experimental results and performance evaluation of the proposed Deepfake Detection and Blockchain Integration system. The system was trained and tested using publicly available datasets such as FaceForensics++, Celeb-DF, and DFDC datasets.

The evaluation process focuses on:

- Deepfake detection accuracy
- Prediction performance
- Blockchain verification reliability
- Real-time processing capability

Performance metrics such as accuracy, precision, recall, F1-score, confusion matrix, and classification reports were used to evaluate the effectiveness of the proposed system.

The chapter also discusses visualization results, deployment performance, and overall findings obtained during system testing.

7.2 Model Performance Metrics

Performance metrics are important for evaluating the quality and reliability of machine learning models.

The proposed system was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix
- Classification report

These metrics help measure prediction correctness and detection capability.

7.2.1 Accuracy

Accuracy represents the percentage of correctly classified predictions.

Accuracy Formula:

Where: True Positives

- TP =
- TN = True Negatives

- FP = False Positives
- FN = False Negatives

The proposed ResNet50 model achieved high prediction accuracy during evaluation.

Observed Accuracy:

- Training Accuracy: 96%
- Validation Accuracy: 94%
- Testing Accuracy: 93%

The high accuracy indicates that the model successfully identifies manipulated media with minimal prediction errors.

The use of transfer learning and preprocessing techniques significantly improved prediction performance.

Precision measures how many predicted positive results are actually correct.

Precision Formula:

$$Pcso = \frac{P}{P + P}$$

Recall measures how effectively the model identifies actual positive samples.

Recall Formula:

$$cll = \frac{P}{P +}$$

The proposed system achieved:

- High precision values
- High recall values
- Reduced false predictions

The model demonstrated strong capability in identifying both REAL and FAKE media samples.

High recall ensures that manipulated content is not missed during detection.

High precision reduces false alarms and improves system reliability.

7.3 Classification Report Analysis

The classification report provides detailed evaluation of model performance for each prediction category.

The report includes:

- Precision
- Recall
- F1-score
- Support values

The model produced strong performance for both:

- REAL class
- FAKE class

The F1-score values remained consistently high, indicating balanced precision and recall performance. The classification report analysis showed:

- Minimal false positives
- Minimal false negatives
- Stable prediction behavior

The use of ResNet50 transfer learning significantly improved feature extraction and classification capability.

The preprocessing techniques such as normalization, face detection, and data augmentation also contributed to improved prediction quality.

The classification report demonstrates that the model generalizes effectively across different datasets and real-world media samples.

7.4 Confusion Matrix Discussion

The confusion matrix visually represents prediction performance.

It contains:

- True Positives

- True Negatives
- False Positives
- False Negatives

The confusion matrix showed:

- High true positive rate
- High true negative rate
- Low misclassification

This indicates that the model successfully distinguishes between genuine and manipulated media.

False positives and false negatives were minimized using:

- Transfer learning
- Fine-tuning
- Data preprocessing
- Balanced dataset training

The confusion matrix confirms that the proposed system provides reliable deepfake detection performance.

7.5 Visualization & Evaluation Insights

Visualization tools were used for analyzing training and prediction behavior.

Visualizations included:

- Accuracy graphs
- Loss graphs
- Confusion matrix heatmaps
- Prediction plots

The training graphs showed:

- Stable convergence
- Reduced overfitting
- Improved learning performance

Loss values decreased consistently during training, indicating proper model optimization.

Accuracy graphs demonstrated gradual improvement in prediction performance over training epochs.

Visualization analysis confirmed that the proposed model achieved stable and efficient learning behavior.

7.6 Real-Time Deployment Results

The developed application was successfully deployed and tested in real-time conditions.

The deployed system successfully:

- Accepted media uploads
- Processed images and videos
- Generated predictions
- Displayed confidence scores
- Generated SHA-256 hashes
- Stored hashes on blockchain
- Verified media authenticity

The frontend interface provided:

- Fast user interaction
- Real-time prediction display
- Secure verification workflow

Blockchain transactions were successfully executed using Ethereum Sepolia Testnet.

The deployment results demonstrated the practical feasibility of integrating Deep Learning and Blockchain Technology into a real-world media verification system.

7.7 Discussion of Findings

The experimental results demonstrate that the proposed system provides:

- High deepfake detection accuracy
- Secure blockchain-based verification
- Tamper-proof authentication
- Reliable prediction performance

The integration of blockchain significantly improved digital trust and media authenticity verification.

The ResNet50 CNN model performed effectively in detecting manipulated media across different datasets.

The use of SHA-256 hashing and Ethereum blockchain ensured secure and immutable verification records.

CHAPTER 8
Conclusions and Future Scope

The proposed Deepfake Detection and Blockchain Integration system successfully combines Deep Learning and Blockchain Technology into a secure and intelligent framework for detecting manipulated digital media and verifying authenticity.

The system uses the ResNet50 Convolutional Neural Network architecture for identifying fake images and videos with high accuracy. Transfer learning, preprocessing techniques, and data augmentation significantly improved model performance and prediction reliability.

The integration of SHA-256 cryptographic hashing and Ethereum blockchain technology provides secure and tamper-proof media verification. Blockchain storage ensures that media authenticity records remain immutable and transparent.

The proposed system successfully achieved:

- Accurate deepfake detection
- Secure blockchain verification
- Reliable media authentication
- Improved digital trust
- Scalable architecture
- User-friendly interaction

The developed framework demonstrates practical integration of Artificial Intelligence, Deep Learning, and Blockchain Technology into a real-world cybersecurity application.

The project can be applied in:

- Journalism
- Social media monitoring
- Digital forensics
- Banking security
- Cybercrime investigation
- Law enforcement systems

The overall results confirm that combining machine learning with blockchain technology provides an effective solution for preventing digital media manipulation and improving authenticity verification.

8.2 Future Scope

Although the proposed system achieved strong performance, several future improvements can further enhance the framework.

Real-Time Deepfake Detection

Future systems can support real-time detection of:

- Live video streams
- Video conferencing platforms
- Social media streaming content

This will improve immediate threat detection and prevention.

Audio Deepfake Detection

Future implementations may include detection of:

- Voice cloning
- Fake speech generation
- Audio manipulation

Audio analysis can improve overall multimedia verification capability.

Mobile Application Development

Mobile applications can be developed using:

- Flutter
- React Native

This will improve accessibility and allow users to perform deepfake verification using smartphones.

Cloud Deployment

Cloud integration using:

- AWS
- Google Cloud

- Microsoft Azure

can improve scalability, storage capacity, and real-time processing performance.

Advanced AI Architectures

Future systems may use advanced models such as:

- EfficientNet
- Vision Transformers (ViT)
- Swin Transformer

These models may further improve detection accuracy and feature extraction capability.

Multi-Blockchain Integration

Future implementations may support:

- Ethereum
- Polygon
- Hyperledger
- Binance Smart Chain

This can improve blockchain flexibility and transaction efficiency.

Social Media Monitoring Systems

The framework can be integrated with social media platforms for:

- Automatic fake media detection
- Content moderation
- Misinformation prevention

This can reduce harmful digital content circulation.

8.3 Summary

The project presented a secure and intelligent framework for detecting manipulated digital media and verifying authenticity using blockchain technology.

The proposed system integrates:

- Deep Learning
- Blockchain Technology

- Cryptographic Hashing
- Web Technologies

Major components of the system include:

- ResNet50-based deepfake detection
- SHA-256 hash generation
- Ethereum blockchain verification
- Flask backend services
- Interactive frontend interface

The system successfully demonstrated:

- High prediction accuracy
- Secure verification
- Tamper-proof authentication
- Scalable deployment capability

The project contributes to research in:

- Artificial Intelligence
- Cybersecurity
- Blockchain Technology
- Digital Forensics

The developed framework provides a practical solution for improving digital trust and preventing manipulation of online media content.

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Chapter 0

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